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Mobile-based deep learning system for early detection of diabetic retinopathy

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ABSTRACT

Diabetic retinopathy (DR) is a leading cause of vision loss globally, especially in regions with limited access to eye care. Early detection is essential to prevent irreversible damage and improve patient outcomes. In this study, a portable, real-time Assisted Mobile Diagnostic (AMD) system for DR detection, which integrates an optimized deep learning model into a mobile platform, is presented. Unlike conventional AI-based approaches that require high-performance computing and stationary fundus cameras, our system combines a non-mydriatic retinal camera with a mobile device, enabling point-of-care diagnostics. Captured retinal images are preprocessed using techniques such as blurring and contrast enhancement before being analyzed by a fine-tuned DenseNet-121 model. The model is trained using a private dataset along with two large public datasets: APTOS (Asia Pacific Tele-Ophthalmology Society) and EyePACS (Eye Picture Archive Communication System). The proposed approach achieved a high accuracy: 97.38% on APTOS, 90.90% on EyePACS, and 98.61% on the private dataset. The system delivers real-time performance on mobile devices, with an average processing time of 162.5 ms, making it well-suited for rapid screening. This Deep learning-based mobile application is part of a multi-platform tele-ophthalmology framework that includes both tablet and desktop integrations, facilitating accessible and remote DR diagnosis.

1. Introduction

Early detection of diabetic retinopathy (DR) is essential to prevent blindness but remains limited by a lack of accessible equipment, particularly in medically under-resourced areas. With the rise of diabetes, which affected 463 million people in 2019 and could reach 700 million by 2025 [1], DR has become a major cause of visual impairment, with a particularly high prevalence in Africa (35.9%), the Middle East and North Africa (32.9%) and North America (33.3%) [2]. Fundus imaging plays a crucial role in the diagnosis of diabetic retinopathy (DR), allowing early signs of the disease to be detected before irreversible complications develop [3,4]. Portable retinal cameras offer a modern alternative to conventional imaging systems, providing a more accessible and adaptable solution for mobile screening settings [5]. Their ability to capture high-quality images in real time and their connectivity with tablets or smartphones facilitate rapid data acquisition and analysis. This technology is particularly beneficial for rural areas where access to ophthalmologists is limited [6]. Integrating an automated diagnostic assistance system would further strengthen this approach, facilitating the identification of high-risk cases and ensuring faster referral to specialists. By opening up access to screening, these new technologies significantly contribute to preventing eye disease and improving the visual health of the population at risk [7].

The research presented in this article is part of a project that aims to design and develop an automatic system to assist in detecting and diagnosing ocular pathologies, using fundus images of the eye captured by portable retinal cameras. However, lesions that can be detected on fundus imaging are caused by the early stages of DR. These lesions appear at an early stage and include microaneurysms, hemorrhages, soft exudates, and hard exudates. Microaneurysms are the earliest signs resulting from the retinal capillaries' weakness. Hemorrhages result from the rupture of microaneurysms, while soft exudates are due to the leakage of lipids from damaged vessels. Hard exudates develop from lipid deposits and signify long-term retinal changes in the advanced stages of DR [8,9]. Early detection enables preventative treatment

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to stop DR progression, thereby reducing the risk of irreparable eye damage. The target system could be deployed on pilot sites as part of Tele-Ophthalmology networks. However, one such deployment is aimed towards the Fez region of Morocco, with a current diabetic patient count of 1.5 million, which is expected to increase to 2.6 million by 2030 [9].

Recent advances in artificial intelligence have demonstrated that automated diagnostic systems using deep learning can achieve high levels of accuracy in medical imaging tasks, including diabetic retinopathy detection [10], and even in more complex applications such as neurodevelopmental disorder classification and brain imaging analysis [11,12]. Therefore, the screening needs to be accessible, calling for a Mobile-Aided Diagnosis (MAD) system, which is mobile, low-cost, and easy to use by an oculist, orthoptist, or nurse. Such a system must ensure the acquisition of patient fundus images in "real-time" using portable retinal cameras (non-mydriatic fundus cameras) connected either to smartphones or digital tablets, which will provide the screening results, thereby assisting in early diagnosis of DR, especially by detecting the earliest stage (i.e. the Mild no proliferative DR). Once the diagnosis results are established during the consultation, a deep examination by ophthalmologists occurs at a remote ophthalmology site [13].

For the intelligent part, we selected DenseNet-121 as the backbone deep learning model due to its proven effectiveness in medical image classification tasks, particularly in retinal disease detection. DenseNet-121 is known for its efficient feature reuse and reduced number of parameters compared to other deep architectures, making it both powerful and computationally efficient. Unlike traditional CNNs (Convolutional Neural Networks), DenseNet's dense connectivity allows better gradient flow and encourages feature propagation, which is especially useful when working with relatively small and imbalanced datasets, such as those used in diabetic retinopathy detection. Preliminary experiments conducted with models such as MobileNet-V2, ResNet50, VGG16, and InceptionV3 showed that DenseNet-121 consistently provided better accuracy and generalization, further supporting its selection in our study.

Our targeted system developed as a Tele-Ophthalmology-based mobile application, will constitute a better approach to providing itinerant visual care in rural areas of Fez (Morocco) and its region (Rural Eye Care) as a first prototype, connected to a pilot site at the Tertiary Ophthalmology Center and Vision Center at Omar Drissi Hospital, CHU HASSAN II, Fez. This will eventually help overcome the serious dearth of ophthalmologists and contribute towards providing healthcare solutions at the patient's doorstep. Currently, only 330 ophthalmologists work in the public sector, whereas more than 80% of the private sector's ophthalmologists are centered on the Casablanca-Kenitra axis [14,15]. Since, the images captured by portable retinal cameras may suffer from limitations such as blurriness, noise, or specific artifacts, which can complicate their interpretation by ophthalmologists and undermine the robustness of a real-time deep learning-based Assisted Mobile Diagnostic (AMD) system for detecting DR, the primary challenge of our approach lies in ensuring the highest possible quality of images acquired through mobile devices. In this context, this work makes the following contributions:

- A comprehensive comparative analysis of noise reduction and contrast enhancement methods adapted for retinal images acquired with portable fundus cameras is presented. The performance of each method is quantitatively assessed using established image quality metrics, PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index Measure), and RCEF (Relative Contrast Enhancement Factor), to determine their effectiveness in enhancing diagnostic image quality under real-world acquisition conditions.
- The optimal preprocessing method is evaluated on two public datasets (APTOS, EyePACS) and then validated under real-world conditions on a private dataset, which was collected from CHU Hassan II Hospital (Fez), ensuring its robustness for the early detection of DR.
- An end-to-end mobile-aided diagnostic (MAD) system is developed, integrating optimized deep learning models (DenseNet-121, Inception-v3, and MobileNetV2) with a tele-ophthalmology application designed for real-time screening and referral. The system has been deployed as a mobile application, facilitating early detection of diabetic retinopathy in underserved and remote areas.

The rest of the paper is organized as follows. Section 2 provides a comprehensive review of existing research on early screening for DR using non-mydriatic handheld fundus cameras and the public datasets of retinal images. Section 3 focuses on our comparative study of fundus image-enhancing methods, covering the datasets used. Section 4 describes the proposed method, which uses deep transfer learning models for early detection of DR, compares its performance with existing approaches, and details the implementation of a mobile application, showcasing its effectiveness for quick and accessible DR detection. Finally, we conclude in Section 5.

2. Related work

There are several levels of severity associated with the disease of DR. The main levels can be categorized as either non-proliferative (NPDR) or proliferative DR, the latter being the advanced stage. NPDR is further categorized into three stages: mild, moderate, and severe [8–16]. For DR detection, numerous computer-based techniques have been developed and reported in the literature. More recently, researchers have started using Deep Learning (DL) techniques for DR detection, particularly the transfer learning (TL) approaches, which have demonstrated promising results for image processing tasks [17]. For example, in [18], authors explored the effect of image enhancement techniques and hyperparameter tuning on DR detection using pre-trained CNN models. By applying CLAHE (contrast-limited adaptive histogram equalization) and using Bayesian optimization on AlexNet, GoogLeNet, and SqueezeNet, they achieved up to 99.81% accuracy in multi-class classification of DR. This demonstrates the importance of preprocessing choices and optimization in improving classification performance. Many computer-aided diagnosis methods for early screening and binary grading of DR have been developed. In [19], the authors have implemented a CNN using standard trainable and non-linear layers for DR detection. Wang et al. [20] explored the use of deep transfer learning models such as AlexNet, VGG-16, and InceptionNetV3 for the diagnosis of DR. Their work demonstrated low accuracy results on retinal images from the Kaggle dataset, with accuracy rates of up to 37.43% for AlexNet, 50.03% for VGG-16 and 63.23% for InceptionNetV3. Pamadi et al. [21] worked using the MobileNetV2 architecture for binary and multimodal classification of DR images, with an accuracy of 78% on the Kaggle APTOS dataset in binary DR detection. Zhou et al. [22] presented a multi-cell architecture designed to increase learning speed and classification accuracy, achieving an accuracy rate of 63.22% for the diagnosis of DR. Deepa et al. [23] presented an ensemble system comprising several deep-learning models to identify and classify DR. The model obtained a classification accuracy of 96.2% with 5-fold cross-validation. Imran Qureshi et al. [24] proposed an active deep learning (ADL) method to identify and categorize the two states of diabetic retinopathy, "DR" and "No DR". It demonstrated a sensitivity of 92.20%, a specificity of 95.10%, an F1-score of 93.1% and an accuracy of 98.1%. Moreover, transfer learning (TL) techniques offer great advantages in achieving high performance while reducing training time, memory requirements, and network design efforts. One of the contributions of this work [25] is an overview of DL algorithm applications to classification-based studies in DR detection using fundus imaging. Moreover, most of the research articles reviewed in [26] focus on the transfer learning DCNN models for DR detection and grading.

To achieve the challenging goal of early detection of DR, the use of portable retinal cameras has become essential to the development of Tele-Ophthalmology, improving the detection of DR [27].

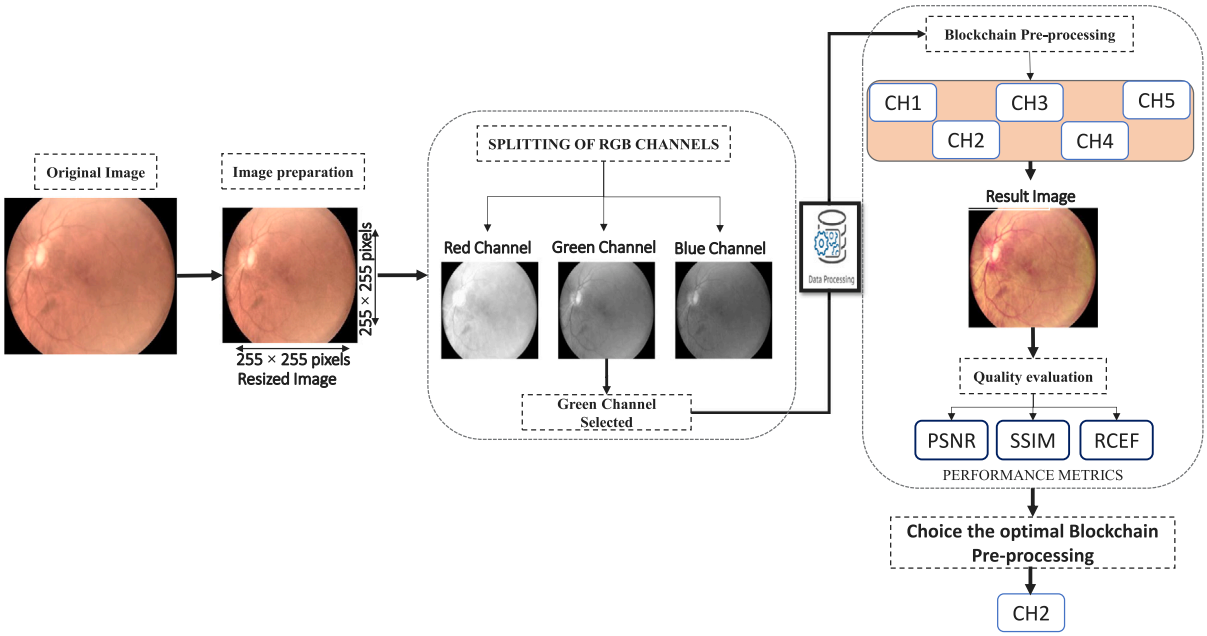


Fig. 1. Flowchart of enhancing retinal image quality through preprocessing block diagram. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Recently, the application of transfer learning with adaptive fine-tuning for DR detection and classification has developed as a key tool in Tele-Ophthalmology [28]. In a study by Medina et al. [29], the Optomed Aurora portable fundus camera outperformed the standard tabletop fundus camera in terms of sensitivity and specificity for DR screening. In another work [30], three portable fundus cameras, iNview, Peek Retina, and Pictor Plus, are compared for their sensitivity and specificity in detecting ocular diseases like DR. These fundus cameras demonstrated good results and effectively detected DR when used for Tele-Ophthalmology retinal screening consultations. The approach proposed in [31] uses a handheld portable fundus camera - Pictor Plus, Volk Optical Inc., USA - and performs DR grading using a deep learning system, Pegasus (AI software, Visulytix Ltd., UK), which resulted in 89.4% and 94.3% detection rates, for RDR and PDR AUROC (Area Under the Receiver Operating Characteristic) curves, respectively. The study was part of the Mexican Advanced Imaging Laboratory for Ocular Research (MAILOR) cohort.

The work presented in this article aims to deploy a telemedicine network for early DR detection. Our ultimate objective is to design and develop a mobile system, including a tele-ophthalmology mobile app that employs Deep Learning models and portable retinal cameras for screening and early diagnosis of DR. We first start with a mobile app for effective early detection of DR, mainly binary detection of “DR” or “No DR” because of its ability to provide rapid and decisive results. This approach enables early and targeted medical intervention, improving patient outcomes by reducing diagnostic delays and facilitating rapid and appropriate treatment. In this context, we are collaborating with the Ophthalmology Center and Vision Center at Omar Drissi Hospital, CHU HASSAN II, Fez [32,33]. Other than DR, our proposed approach can be used and easily adapted to other eye diseases, as shown by our work on Glaucoma [34] and AMD (age-related macular degeneration) [35].

3. Comparative analysis of fundus image enhancement techniques

Retinal fundus images captured with handheld cameras often suffer from quality problems such as blurring, low contrast, noise, and uneven illumination, particularly in low-resource or outdoor environments. These impairments significantly hamper the visibility of key features such as microaneurysms, hemorrhages, and exudates, which are essential for the diagnosis of DR at an early stage [36,37]. Recent studies have highlighted the importance of pre-processing to improve image quality before classification. For example, contrast enhancement techniques have been shown to improve lesion visibility and, consequently, the performance of deep learning models in DR detection [38,39]. Furthermore, the use of adaptive enhancement pipelines combining denoising and CLAHE has demonstrated substantial gains in both model sensitivity and specificity. Therefore, the design of a robust and lightweight image enhancement pipeline is essential to guarantee the diagnostic quality of fundus images captured in the field.

In this study, we perform a comparative analysis of several preprocessing pipelines designed to enhance retinal images captured during real-world mobile screening. Our approach emphasizes the selection and evaluation of combinations of denoising, contrast enhancement, and adaptive gamma correction filters, aiming to identify the most effective configuration for improving diagnostic image quality. The chosen pipeline focuses on the green channel to preserve vascular details, and its performance is assessed using quantitative metrics such as PSNR, SSIM, and RCEF. The overall image enhancement framework is illustrated in Fig. 1.

The proposed preprocessing for enhancing retinal fundus images includes several steps. The retinal image is first captured and resized to optimize its dimensions, thereby reducing processing time. Next, the image is partitioned into distinct color components, namely red, green, and blue. Emphasizing the green channel assumes critical importance due to its ability to preserve intricate color details within the image [28]. Subsequent image enhancement is achieved through the application of various filters, including sequential combinations of smoothing and sharpening techniques, to further refine the enhanced image. Finally, the effectiveness of the entire process is evaluated using three metrics: the PSNR, the SSIM, and the RCEF. These metrics offer valuable insights into the final image quality and serve as a quantitative assessment of the performance at each stage of the process.

3.1. Private dataset description

3.1.1. Dataset collection process

In the field of DR research, several projects have recently focused on creating private databases using portable retinal cameras to better

Fig. 2. oDocs nun IR retinal camera [40].

understand and analyze this condition. Collecting data at the local level through these portable technologies improves detection systems by adapting them to regional characteristics and demographic differences among patients [28,30]. Therefore, collecting private datasets is essential, as it provides more relevant data adapted to each context. Our collaboration with the Department of Ophthalmology at Omar Drissi Hospital (CHU Hassan II) in Fez, Morocco, emphasizes this approach by using a portable retinal camera, which allows for efficient data collection for more accurate and personalized DR detection.

The private dataset collected comprises retinal fundus images obtained using the oDocs Nun IR camera, a portable and non-mydriatic imaging device specifically designed for DR detection, as presented in Fig. 2. To ensure representativeness and clinical relevance, images were collected from patients diagnosed with varying stages of DR, as confirmed through ophthalmological examinations. The dataset collection followed strict inclusion criteria, targeting patients aged 18 years and older with a confirmed history of diabetes. Additionally, patients with other retinal pathologies or ocular conditions were excluded to maintain the specificity of the dataset for DR analysis.

Compared to public datasets such as APTOS and EyePACS, which are typically collected in clinical environments using high-end retinal cameras under standardized conditions, our private dataset provides greater variability and realism. The images were captured at Omar Drissi Hospital (CHU Hassan II) in Fez, Morocco, using a handheld retinal camera during mobile screening campaigns targeting patients from rural and semi-urban areas. As a result, the dataset exhibits a wider range of lighting conditions, image quality, and capture angles, reflecting real-world challenges that are often overlooked in benchmark datasets. Additionally, the patient population in our dataset is demographically diverse, including individuals from various age groups, socio-economic backgrounds, and ethnicities, without any preferential selection or filtering, ensuring a representative sample reflective of the Moroccan context. This variability enhances the robustness and adaptability of the proposed detection framework to underrepresented and low-resource settings.

The choice of the oDocs Nun IR retinal camera for image acquisition is based on several key advantages. It offers a high resolution of 2880 × 2160 pixels and a wide field of view (FOV) of 45° to 55°, allowing for precise visualization of retinal features. Its manual focus capability, ranging from −20D to +20D, ensures optimal image sharpness. The camera's portability and compatibility with Android smartphones make it ideal for use in remote areas with limited ophthalmological resources.

The oDocs Nun IR captures critical details such as microaneurysms and hemorrhages, essential for accurate detection and staging of diabetic retinopathy (DR). This device combines high performance with accessibility, particularly in hard-to-reach areas. Additionally, the proposed pipeline is compatible with other portable retinal cameras such as Vistaview, Volk iNview, and Peek Retina, providing flexibility for various clinical contexts and ensuring effective detection of DR across different environments. The system was developed with modular components for image acquisition, preprocessing, and inference, which facilitates its adaptation to new devices with minimal adjustments. This modularity supports scalability, allowing seamless integration of other mobile smartphones or stationary fundus cameras with varying image formats and specifications.

3.1.2. Data quality and preprocessing

The private dataset used in this work consists of 1319 retinal images, exhibiting a range of quality attributes. Fig. 3 illustrates representative samples from the dataset. High-quality images (Fig. 3.a) demonstrate precise details, minimal noise, and enhanced contrast, ensuring their suitability for analysis. Conversely, the dataset includes images with notable quality deficiencies, such as blur (Fig. 3.b), noise (Fig. 3.c), and insufficient contrast enhancement (Fig. 3.d). To address these issues, additional image processing techniques, including deblurring and contrast enhancement, were applied to ensure that all images met the necessary standards for ophthalmological interpretation and deep learning analysis.

3.1.3. Dataset annotation and labeling

The data collection process strictly adhered to ethical guidelines to ensure patient confidentiality and compliance with international data protection standards. Ethical approval was obtained from an independent ethics committee, and all participants provided written informed consent before data collection. To safeguard patient data, robust security measures were implemented, including image and metadata anonymization, secure storage on encrypted servers, and restricted access limited to authorized personnel. These measures ensured that the dataset adhered to high standards of data privacy and security, following best practices in medical research. Nevertheless, these devices can only be used under the full responsibility of healthcare professionals, even in remote areas, to ensure that ethical aspects and patient data protection are taken into account. A specific procedure will then be set up in consultation, which will validate its use as an early diagnosis tool.

The dataset was annotated by a team of three experienced ophthalmologists, who independently reviewed and classified first the images into five distinct stages of DR: absence of DR, non-proliferative diabetic retinopathy (NPDR) minimal, NPDR moderate, NPDR severe, and proliferative diabetic retinopathy (PDR). To ensure inter-rater reliability, classification disagreements were resolved through a consensus process, involving a fourth expert when necessary. The final labeled dataset is distributed as follows: 256 images with no DR, 381 images with minimal NPDR, 335 images with moderate NPDR, 253 images with severe NPDR, and 94 images with PDR. The labeling process adhered to internationally recognized DR classification guidelines, ensuring consistency, accuracy, and reproducibility in the dataset. Moreover, the "No DR" stage was separated, while the four stages containing retinal lesions were merged into a single stage labeled "DR". This approach facilitates the identification of DR in its early stages while simplifying the classification of more advanced forms of the disease.

3.2. Public datasets description

In addition to the private dataset used in this study, two widely recognized public datasets, APTOS [41] and EyePACS [42], were also employed. The increasing demand for validation and training of deep learning models has driven research groups to create and release exclusive datasets to the public [43]. Among these, the APTOS and EyePACS datasets have gained significant recognition in the research community for their comprehensive collection of high-resolution retinal fundus images.

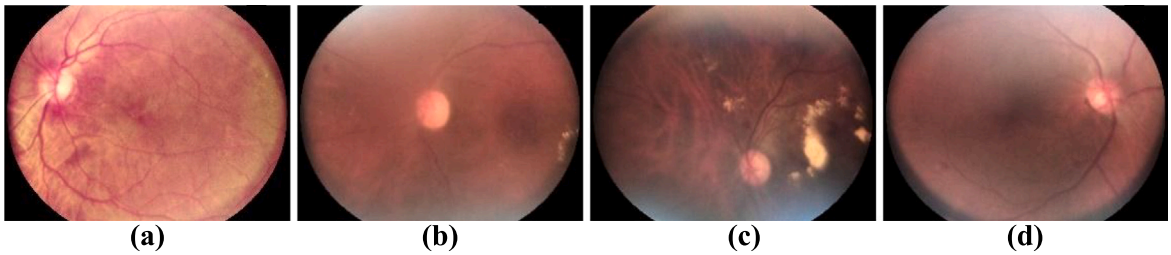


Fig. 3. Example of image clarity (a), Blurriness (b), Contrast (c), and Brightness variation (d) from the private dataset.

3.2.1. APTOS

The APTOS [41] Blindness Detection dataset proves to be a valuable resource for researchers and practitioners in the field of DR. This dataset encompasses a collection of fundus images effectively portraying the five distinct stages of DR. Each image is thoroughly labeled with a severity level, which spans the spectrum from 0 to 4. These severity levels correspond to specific categories: Label 0 signifies the absence of DR, label 1 indicates mild DR, label 2 represents moderate DR, label 3 denotes severe DR, and label 4 indicates proliferative DR. In total, the dataset consists of 3662 retinal images. Among these, 1805 images are meticulously classified as exhibiting no DR, 370 images are labeled as mild DR, 999 images are classified as moderate DR, 193 images are identified as severe DR, and 295 embody the characteristics of proliferative DR. The fundus image resolution within this dataset maintains a high standard at 3216 × 2136 pixels, ensuring a repository of visuals with remarkable quality, optimally suited for analysis and evaluation endeavors.

3.2.2. EyePACS

The EyePACS [42] dataset, similar to the APTOS dataset, provides an extensive compilation of fundus images capturing the complete spectrum of DR progression across its five stages. Comprising a total of 35 126 retina images, with each image corresponding to both the left and right eyes. These images boast dimensions of 3888 × 2951 pixels, ensuring high-resolution visual data for analysis and research purposes. Within the EyePACS dataset, the images are meticulously categorized according to the five DR stages. Specifically, there are 25 810 images labeled as having no DR, 2443 images classified as mild DR, 5292 images portraying moderate DR, 873 images indicating severe DR, and 708 images showcasing proliferative DR. For the training process of the proposed model, a random selection strategy is adopted, choosing 1534 images from the no DR stage and 2764 images from the DR stage.

3.3. Proposed flowchart and analysis of enhancement techniques

Image quality enhancement techniques aim to optimize images by reducing undesired disturbances, enhancing the visibility of important features, and improving overall picture quality. One effective approach for enhancing image quality involves accentuating the contrast between dark and light regions. This can be achieved through the application of filters, which are commonly employed in low-quality image processing tasks. A variety of filters can be used to enhance the visual quality of images, including those that provide a softer appearance, enhance details, or emphasize edges. Previous research has explored the effectiveness of employing 11 simple filters to enhance the quality of retinal images [39]. These filters have demonstrated reliability in improving the clarity of blood vessels within the images. Before applying these filters, the images undergo initial preprocessing steps, such as resizing and selection of the "Green channel", which contains the most detailed color information. In our approach, we specifically employ 5 high-performance filters that have yielded favorable outcomes. These filters include the median filter (MF), the non-local means filter (NLMF), the Gaussian filter (GF), CLAHE, and the bilateral filter (BF). Detailed descriptions of these filters, along with their respective mathematical formulas, are provided, and the resulting images they produce are visually illustrated in Table 1. In this context, the median filter is applied as a noise-reduction technique that preserves edges by replacing pixels with the median value of their neighboring pixels. This method is particularly beneficial for retinal images, as it effectively removes extreme values caused by noise. On the other hand, CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances image contrast by adaptively adjusting pixel values based on local regions. This approach is especially effective in highlighting critical features such as blood vessels, although careful application is required to prevent the amplification of noise. Both filters play a crucial role in improving the quality and analyzability of retinal images. In addition to the filters mentioned above, we also implemented and evaluated the enhanced median filter (EMF), as well as the median filter combined with an average filter (MMF) [44]. It is noteworthy that the MMF performs poorly compared to the other five tested filters in terms of PSNR, SSIM, and RCEF [39], and these results are not presented here.

3.3.1. Enhancement chains

To construct preprocessing chains, the most successful filter outcomes are selected. These chains consist of multiple stages, including denoising, contrast enhancement, and the application of adaptive gamma correction. Among the filters that demonstrated notable effectiveness are those listed in Table 1. The MF, GF, BF, and NLMF work as smoothing filters, while CLAHE acts as a sharpening filter. As a result, a combination of both smoothing and sharpening filters is employed within the chains. Adaptive gamma correction (AGC) is a technique that adjusts the intensity values of pixels in an image based on a gamma correction curve (GCC) [45]. According to this formula, for an image of size m × n pixels, the gamma correction transforms the intensity value of the pixel at each spatial position (x, y) according to the following expression:

$$I_{\text{gam}}\ (x, y) = I(x, y)^{\gamma} \tag{1}$$

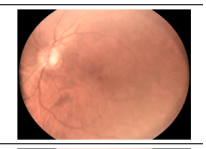
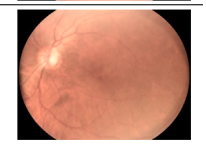
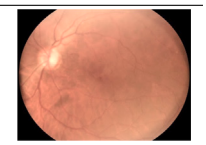
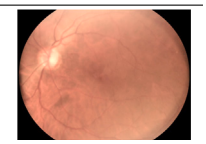
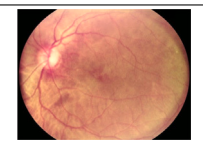
where $I_{\text{gam}}\ (x, y)$ represents the gamma-corrected intensity of the image I (x, y), and γ represents different gamma values that control the type of mapping. When γ is set to a value $\gamma > 1$, the dark regions of the original image become brighter in the gamma-corrected intensity image. Conversely, when γ is set to a value $\gamma < 1$, the bright regions of the original image become darker in the gamma-corrected intensity image [38]. It is important to note that the gamma value is adaptive for each image, which means that each image has its gamma value based on its intensity characteristics.

3.3.2. Performance metrics

The performance of the image processing algorithms or enhancement filters can be evaluated by various quantitative measures. The metrics, such as PSNR, RCEF, and SSIM, are used to find which image enhancement techniques give better results. PSNR is the most commonly used metric for image enhancement techniques and is given by:

$$PNSR = 10\log_{10}\frac{255^2}{\frac{1}{MN}\sum_{ij}\left[g_{ij} - f_{ij}\right]^2}dB \tag{2}$$

Table 1
Filters with their mathematical expressions and output images.

Filters	Mathematical expressions	Output images
MF	$g(x,y) = \text{median}\{f(n,m)\}$ Where $f(n,m)$ returns the pixel value at position (n,m) in the image.	
GF	$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma}$ Where G(x, y) is the value of the Gaussian filter at a given position (x, y). σ: the standard deviation of the Gaussian filter.	
BF	$I = \frac{1}{W_P} \sum_{X_i} I(x_i) f_i \left(\left\lVert I(x_i) - I(x) \right\rVert\right)$ Where w_P is the normalization factor. $I(x)$ is the intensity of the pixel being processed. $\mathbf{x_i}$ are the neighboring pixels around the pixel being processed.	
NLMF	$\text{NLMF}(x) = \left(\frac{1}{Z(x)}\right) * \sum(w(x,y) * f(y))$ Where Z(x) is a normalization factor. $\sum(w(x,y) * f(y))$ represents the weighted sum of similar pixel values f(y) based on their weights w(x, y).	
CLAHE	$g = [g_{\max} - g_{\min}] * p(f) + g_{\min}$ Where $g_{\max}, g_{\min}$ are the maximum and minimum grey levels. $p(f)$ is the cumulative probability of grey level f in a local region of the input image.	

where f and g are the original and the enhanced images, and M and N are the row and column pixels in the image, respectively. The subscripts i and j correspond respectively to the row and column indexes. The highest PSNR value gives better performance. The change in the structural similarity between the original and enhanced images is measured by the SSIM and is given by:

$$\text{SSIM}(A,B) = \frac{(2\mu_A\mu_B + c_1)(2\sigma_{AB} + c_2)}{(\mu_A^2 + \mu_B^2 + c_2)(\sigma_A^2 + \sigma_B^2 + c_2)} \tag{3}$$

where μ_A and μ_B are the averages of the intensities in the images x and y, respectively. These averages are calculated over a local neighborhood or window of pixels. ${\sigma_A}^2, {\sigma_B}^2$ and σ_{AB} are the variance and covariance of x and y, and c_1 and c_2 are constants added to the formula and chosen to avoid division by zero and ensure numerical stability. The SSIM values range from −1 to 1; if the value is exactly one, it is considered to be of a perfect structure similarity, and if the value is lower, then there is no structural similarity.

The dynamic range of the histogram is measured using RCEF and is given by:

$$\text{RCEF} = \frac{\sigma_\text{B}^2/\mu_\text{B}}{\sigma_\text{A}^2/\mu_\text{A}} \tag{4}$$

where σ_A and σ_B are the standard deviation of the original and enhanced image, and μ_A and μ_B are the mean of the pixel values in the original and enhanced images, respectively. The mean provides a measure of the central tendency or average intensity of the image. If RCEF is greater than 1, it suggests that the enhanced image has a higher contrast and a wider range of pixel values compared to the original image.

Although enhancement techniques improve overall image clarity, excessive sharpening or filtering may amplify noise or distort small pathological features. We therefore used objective performance metrics (PSNR, SSIM, and RCEF) to select the preprocessing chain least likely to cause loss of diagnostic information, particularly early signs of diabetic retinopathy.

Table 2
Performance evaluation of image quality using PSNR, SSIM, and RCEF metrics.

	Blockchain	SSIM	PSNR	RCEF
Chain **1**	MF+CLAHE+AGC	0,9804	33,2531	1,0177
Chain 2	**GF+CLAHE+AGC**	**0,9808**	**33,2950**	**1,0476**
Chain 3	BF+CLAHE+AGC	0,9803	33,2317	1,0378
Chain 4	NLM+CLAHE+AGC	0,9804	33,2096	1,0381
Chain 5	MMF+CLAHE+AGC	0,9802	33,2641	1,0276

3.3.3. Image quality evaluation

The five image enhancement chains, thus proposed, are separately evaluated, and the results are shown in Table 2. Significant findings regarding the different filter chains are highlighted. Among them, this chain, which incorporates the GF+CLAHE+AGC filter combination, stands out as it demonstrates superior performance compared to the others. Chain 2 achieves a high SSIM value of 0.9808 and a PSNR value of 33.2950, indicating a closer structural similarity between the reference and improved image quality. Due to its superior performance, Chain 2 was selected for use in the subsequent stages of the detection and classification approach discussed in Section 4. This preprocessing pipeline is later integrated into our classification framework, where it contributes to improved model accuracy across all datasets. Fig. 4 showcases the outcomes of the best optimal preprocessing blockchain, yielding the respective output images for (a) and (b), denoted as (c) and (d) in the sequence. The processing chain involves the application of a Gaussian Filter followed by merging channels. The Gaussian Filter helps in reducing noise and smoothing the images, while the merging channels operation combines the different color channels to create the final enhanced output. This procedure aims to enhance the quality of retinal images, enabling improved analysis and interpretation for various applications.

The early screening for DR involves two distinct stages: non-proliferative DR (NPDR) and proliferative DR (PDR). NPDR represents the initial stage of the disease, characterized by retinal lesions such as

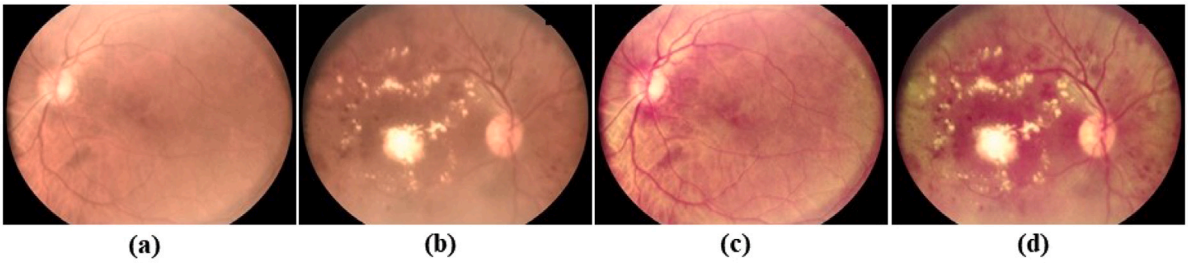


Fig. 4. The best enhancement chain involves Gaussian Filter, CLAHE, and Gamma correction to improve retinal image quality (a) and (b) original images, (c) and (d) are the output results corresponding to (a) and (b), respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

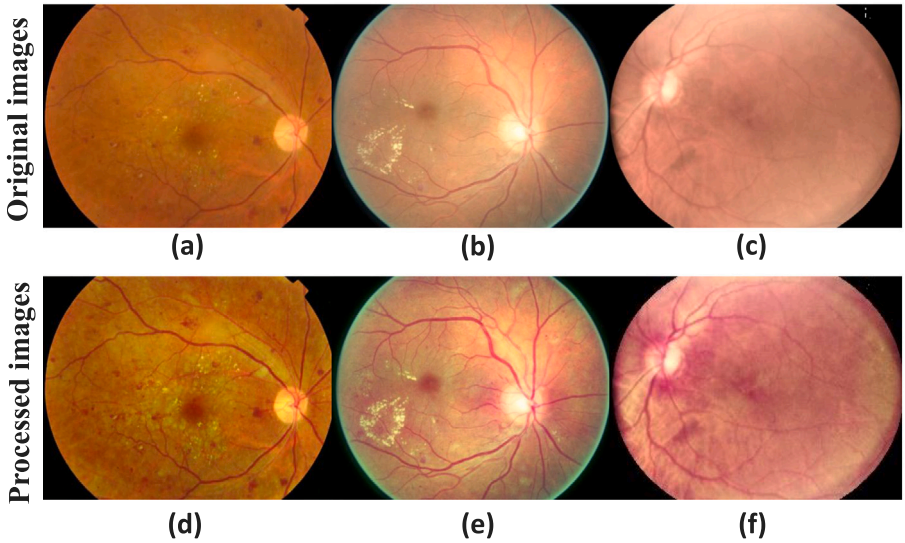


Fig. 5. Preprocessed retinal fundus images: (a) and (d) APTOS dataset, (b) and (e) EyePACS dataset, (c) and (f) private dataset.

microaneurysms, hemorrhages, and exudates. On the other hand, PDR corresponds to the advanced stage, characterized by the formation of new abnormal blood vessels in the retina [46]. The acquired images enable the visualization of characteristic DR lesions and facilitate the determination of the disease stage [47]. In this study, a combination of the three non-proliferative diabetic retinopathy (NPDR) stages (minime, moderate, severe) and the PDR into a single class named "DR" is considered, while keeping the "No DR" class separate. Fig. 5 shows the original images from the private dataset (a), APTOS (b), and EyePACS (c) respectively, and the corresponding results of the optimal preprocessing blocks (d), (e) and (f) used in this research.

As the image quality improvement is a targeted objective, we also decided to check which model architecture best complements the image enhancement pipeline. The global was to analyze how different enhancement techniques affect classification performance across various models and to determine a robust combination suitable for real-world deployment. This approach bridges both key challenges in DR diagnosis: poor image quality and reliable classification under real-world variability.

4. Proposed method

Image classification consists of assigning a specific class or label to a new, unseen image. This process leverages pre-trained models to predict class labels for unknown images based on a dataset of annotated examples. Deep neural networks (DNNs) are among the most widely used architectures for image classification due to their ability to automatically learn hierarchical features such as edges, textures, shapes, and their spatial relationships. These networks are highly proficient at detecting and encoding intricate patterns within data, making them particularly effective for tasks that require high accuracy in image recognition and classification. The method proposed in this study comprises three essential steps. First, data preparation is performed to prepare the model for the classification task. Second, data pre-processing is performed by selecting the best pre-processing chain identified in the study mentioned in Section 3. In addition, data augmentation is applied to increase data diversity and improve model generalization, while the cross-validation 5-Fold technique is used to robustly evaluate model performance. However, three deep learning models, based on the transfer learning approach and fine-tuning of the DenseNet-121, Inception-v3, and MobileNet-v2 architectures, are employed. These pre-trained models are further refined to extract a wide range of lesion features from retinal images, enhancing the model's ability to identify and classify subtle patterns indicative of diabetic retinopathy. Finally, the early detection of diabetic retinopathy has been implemented into a mobile application designed to detect the presence of DR. The detailed flowchart of the proposed method is shown in Fig. 6.

The pre-trained architectures were selected specifically to optimize the detection of the two stages of diabetic retinopathy, "NO DR" or "DR", by highlighting specific features extracted at different levels and taking into account the particular structure of the lesions characterizing DR. To achieve this, DenseNet-121, known for its dense structure, enables efficient communication between the different layers of the network. This architecture facilitates the extraction of fine and complex features present in retinal images, highlighting the subtle details of lesions characteristic of DR, such as microaneurysms, exudates, and hemorrhages. The case is similar for Inception-v3, which features an

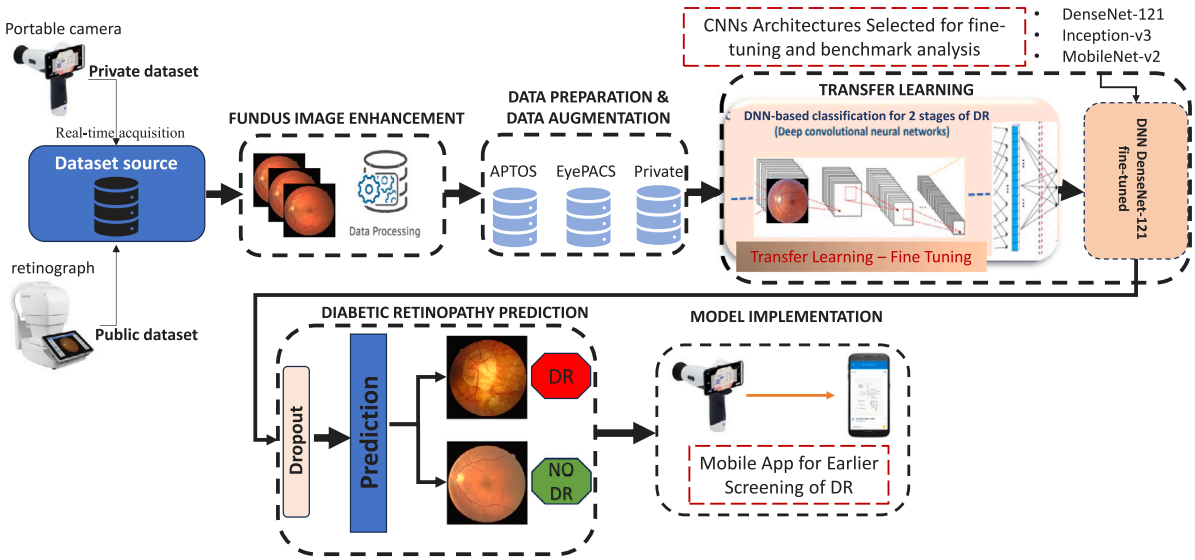


Fig. 6. Flowchart of a deep learning-based mobile app for early screening of diabetic retinopathy using real-time handheld fundus imaging and validation with public datasets.

Table 3
Dataset distribution before and after data augmentation.

Datasets	Before data augmentation		After data augmentation
	Initial dataset	Selected images	Used images
Private	1321	1321	7200
APTOS	3662	2880	9566
EyePACS	35 126	4290	20 000

Inception module that facilitates feature extraction at different spatial scales. This capability is particularly useful for capturing the complex structures of DR lesions, highlighting blood vessel deformations, areas of vascular ischemia, and other characteristic alterations observed in images of the retina affected by the disease. On the other hand, MobileNet-v2, which is designed to maintain high performance while being adapted to the constraints of mobile devices, excels in extracting relevant features with a lightweight architecture. This capability is crucial for highlighting the distinctive features of retinal lesions while ensuring efficient execution on mobile platforms and facilitating the detection of DR stages.

4.1. Data preparation

For the three datasets used in DR detection, each dataset was divided into two subgroups: “NO DR” and “DR”. The “NO DR” subgroup consists of images without any signs of diabetic retinopathy, while the “DR” subgroup includes images from all four stages of DR severity (1, 2, 3, and 4). This categorization facilitates the distinction between images without retinopathy and those exhibiting varying levels of severity, thereby enabling effective analysis and classification of DR cases. To address challenges related to poor image quality and anomalies in public datasets, certain images containing artifacts were excluded from the APTOS and EyePACS datasets. This exclusion was necessary due to saturation problems, which could negatively impact system performance. The objective of this process was to ensure that only high-quality images were retained for further analysis.

4.2. Data augmentation and data splitting

In the preprocessing step, the image resizing was performed to adjust the image size from the preprocessed 255 × 255 pixels to the required dimensions of 224 × 224 pixels, ensuring compatibility with the DNN model’s input. Subsequently, a variety of data augmentation techniques were applied to enrich the dataset. These techniques encompassed vertical and horizontal flipping, rotation at various angles, and adjustments to brightness and color settings. However, the implementation of data augmentation has been seen as a balanced factor at each stage to mitigate the imbalance between the “NO DR” and “DR” classes. By incorporating these augmentation techniques and ensuring that a balance is maintained between the “DR” class and the “NO DR” class, we aim to enrich the training data by introducing more diversity and variability. This step aims to enhance the model’s ability to generalize effectively and capture the wide range of possible scenarios, thus improving its robustness and its ability to handle different cases accurately. Table 3 outlines the distribution and number of images selected for each dataset, along with the count of images before and after data augmentation.

Several studies have employed various proportions to divide datasets into training, validation, and test sets. The allocation of data into different subsets, namely training, validation, and testing sets, holds significant importance in optimizing the learning process and assessing model performance and generalization capabilities. In this context, the distribution of data emphasizes allocating a larger portion to the training set, which facilitates more effective learning by exposing the model to a diverse range of examples. This distribution strategy strikes a balance between maximizing the training data available to the model for effective learning and guaranteeing sufficient data for rigorous evaluation of performance and generalization potential. A common approach allocates 80% of the data for training, 10% for validation, and the remaining 10% for testing [48,49]. Alternatively, some studies have chosen to assign 75% of the data for training, with the remaining portion divided between validation and testing [50]. In the specific case under consideration, a distribution of 75% for training, 15% for validation, and 10% for testing has been selected across all three datasets based on the number of images in each dataset. The images for the private dataset of 7200 images were allocated as follows: 5400 images for training, 1080 images for validation, and 720 images for testing. For the APTOS dataset of 9566 images, 7174 images were selected for training, 1433 for validation, and 959 for testing. Finally, for the EyePACS dataset containing 20,000 images, a similar distribution was applied: 15,000 images were assigned to training, 3000 to validation, and 2000 to testing.

To mitigate class imbalance and enhance the model’s sensitivity to all DR stages, we applied targeted data augmentation to underrepresented classes within the private dataset. This strategy allowed us to obtain a balanced training distribution across the five DR severity

levels, ensuring equitable representation during training and reducing potential bias toward the majority class, particularly the NoDR class, which naturally dominates the raw dataset. Moreover, to prevent information leakage and ensure an unbiased evaluation, data augmentation was applied exclusively to the training set after partitioning the dataset into training, validation, and testing subsets. This approach guarantees that no augmented images appear in the validation or test sets, preserving the integrity of the evaluation process and preventing overestimation of the model's performance due to data redundancy.

4.3. Designing DNN model for earlier screening of DR

To develop a Deep Neural Network (DNN) model for the binary detection of DR, an approach employing transfer learning with three unique architectures is incorporated as a separated bloc in the proposed flowchart (Fig. 6): DenseNet-121 [51], Inception-v3 [52], and MobileNetV2 [53]. The primary objective of this study is to accurately classify two stages of DR: the presence or absence of DR. Through transfer learning with these three architectures, the extensive knowledge and insights gained from large-scale image datasets are leveraged. This approach significantly enhances the ability to differentiate the early stages of DR, particularly the presence of DR for a subsequent initial examination, by effectively leveraging the valuable information encapsulated in pre-trained models.

The developed architectures undergo training for 200 epochs using the RMSProp (Root Mean Square Propagation) optimization algorithm and the Binary Cross-Entropy (BCE) loss function. The choice of RMSProp as the optimization algorithm and BCE as the loss function is justified by their effectiveness in training the developed architectures over 200 epochs. The learning rate is set to 0.001, and a fixed batch size of 64 is used for training and 32 for validation. In addition, we used the 'sigmoid' activation function in the final layer of the DNN model. This function is commonly employed in binary classification tasks as it produces output from 0 to 1, which is suitable for our classification problem. However, to mitigate the risk of overfitting, the early stopping technique is incorporated, where training is halted after 50 epochs if there is no improvement in validation accuracy. Furthermore, a dropout layer is introduced to the model to address overfitting concerns further. The validation accuracy is evaluated at each epoch, and the model with the highest validation accuracy is saved using the model checkpoint functionality of the Keras callback mechanism.

4.4. Performance results and discussion

4.4.1. Implementation framework

The outlined experimentation in this study was conducted on a computer system featuring an NVIDIA Corporation GP102GL [Quadro P6000]. The computer was equipped with an Intel (R) Xeon(R) Gold 6240R, 20 core with 2,40 GHz, and boasted a substantial RAM capacity of 377 GB. The implementation of the models was performed using various software packages, including Python 3.10.9, and deep learning libraries such as Keras with 3.10.9, Keras with Tensorflow, OpenCV, and Scikit-Learn were utilized in conjunction with the aforementioned libraries for the experimentations.

4.4.2. Evaluation metrics

Performance evaluations rely on five commonly used metrics, namely, accuracy, sensitivity, specificity, precision, and F1-score, as shown in Eqs. (5), (6), (7), (8), and (9), respectively. These metrics provide insights into different aspects of the classification model's performance.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{5}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

Table 4
Comparative analysis of performance metrics for three architectures on Private, APTOS, and EyePACS Datasets.

Dataset	Test prediction	Architectures		
	Eval. metrics	DenseNet-12	MobileNetv2	Inception-v3
Private	Acc	**98.61**	96.11	96.52
	Sen	**98.33**	94.72	94.72
	Spe	**100.00**	99.72	99.72
	Pre	**98.88**	98.27	98.27
	F1-score	**70.03**	66.97	66.97
APTOS	Acc	**98.03**	97.49	97.38
	Sens	**97.76**	96.42	97.09
	Spe	**99.78**	100.00	100.00
	Pre	**98.20**	98.40	97.53
	F1-score	**67.98**	70.21	65.78
EyesPACS	Acc	**90.90**	87.85	89.14
	Sens	**93.90**	85.30	88.10
	Spe	**99.40**	98.90	98.90
	Pre	**88.58**	89.88	89.89
	F1-score	**69.37**	70.00	68.37

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{7}$$

$$\text{Precision} = \frac{TP}{\text{TP} + FP} \tag{8}$$

$$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{sensitivity}}{\text{Precision} + \text{sensitivity}} \tag{9}$$

4.5. Results and discussion

Fine-tuning includes adapting a pre-trained model to detect the early stages of diabetic retinopathy by fine-tuning its weights on a smaller, task-specific dataset. This approach leverages a model initially trained on a general task and specializes in recognizing the unique features associated with DR detection. Fine-tuning not only improves the model's ability to generalize but also reduces the need for large amounts of labeled data, making it particularly effective in medical imaging, where annotated datasets can be limited.

The classification results were evaluated using the datasets described in Section 3.1.3 to assess the performance of the proposed system. As summarized in Table 4, the performance metrics demonstrate the superiority of DenseNet-121 across all datasets. DenseNet-121 achieved an impressive accuracy of 98.61%, a sensitivity of 98.33%, a specificity of 100.00%, a precision of 98.88%, and an F1-score of 70.03%. The results obtained on the APTOS and EyePACS datasets are particularly impressive, where DenseNet-121 achieved accuracy rates of 97.38% and 90.90% respectively. These findings confirm the effectiveness of DenseNet-121 in accurate DR detection, showing its capacity to generalize well across diverse datasets, even when faced with variations in data quality and patient demographics.

Further analysis, presented in Table 5, compares the training and validation accuracy for each deep learning architecture across the three datasets. DenseNet-121 consistently outperforms the other models, including Inception-v3 and MobileNetV2, demonstrating its robustness in handling both training and validation phases. This architecture's superior performance can be attributed to its dense connectivity mechanism, which enhances feature propagation and reusability, allowing the model to learn more complex patterns and achieve more accurate predictions. The consistent improvement in performance across all stages further reinforces the reliability of DenseNet-121 for complex classification tasks.

Our findings indicate that DenseNet-121's strong performance can be effectively leveraged not only in advanced medical diagnostics but also in mobile environments where computational efficiency and real-time inference are crucial. This is particularly important in teleophthalmology, where deploying such models on handheld devices can

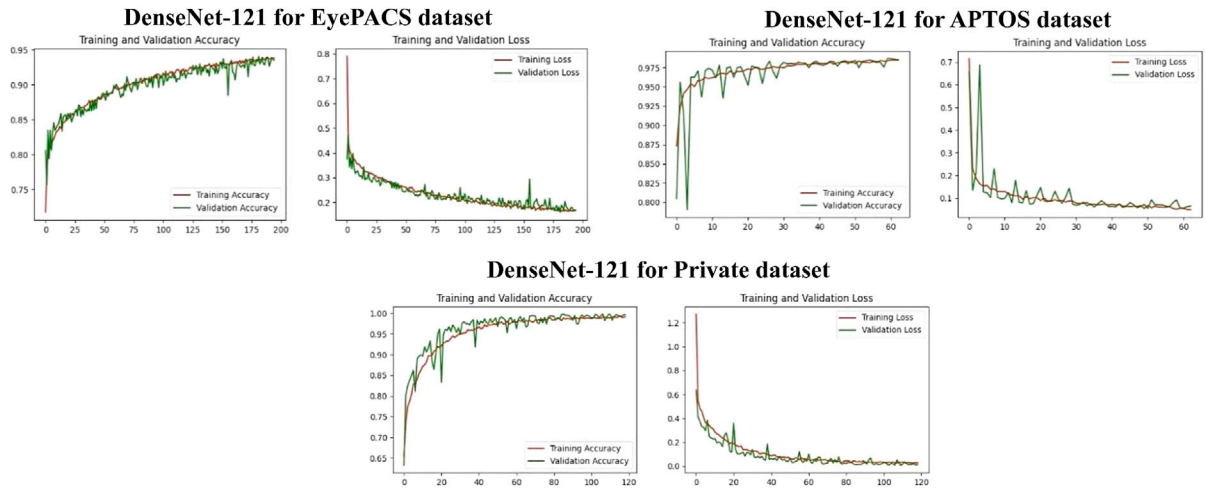


Fig. 7. Accuracy and loss in training and validation for DenseNet-121 architecture on (a) private dataset, (b) APTOS dataset, and (c) EyePACS dataset.

Table 5
Training and validation results comparison of the three architectures.

Dataset	DenseNet-121		MobileNet2		Inception-v3	
	Training	Validation	Training	Validation	Training	Validation
Private	**99.81**	**99.37**	99.07	98.12	98.88	98.42
APTOS	**98.91**	**98.81**	98.68	98.51	98.61	98.60
EyePACS	**94.06**	**93.92**	94.54	92.16	91.30	91.43

Table 6
Confusion matrix of DenseNet-121 architecture on the Private, APTOS, and EyePACS dataset.

		DenseNet-121 prediction					
		Predicted values					
		Private dataset		APTOS dataset		EyePACS dataset	
		NO DR	DR	NO DR	DR	NO DR	DR
True values	NO DR	**356**	4	**459**	11	**879**	121
	DR	6	**354**	13	**435**	61	**939**

enable fast and accessible diabetic retinopathy (DR) screening. While models such as Inception-v3 and MobileNetV2 are also widely used, DenseNet-121 demonstrates a more favorable balance between accuracy and resource demands. Specifically, its dense connectivity pattern enables efficient feature reuse, reducing the number of parameters and mitigating overfitting, contributing to both higher accuracy and better generalization.

In addition, the preprocessing steps played a pivotal role in enhancing classification performance. For DenseNet-121, which benefits from deep and densely connected layers, these preprocessing techniques improve the signal-to-noise ratio and emphasize critical retinal features, thereby facilitating the extraction of discriminative patterns across hierarchical layers. This synergy between preprocessing and architectural design allows the model to capture subtle pathological cues more effectively than shallower or less tightly connected networks. Moreover, compared to lighter models like MobileNetV2, DenseNet-121 achieves significantly improved classification performance while still maintaining acceptable inference times for practical deployment on modern mobile hardware. Moreover, its deeper architecture allows for capturing complex retinal features that may be missed by simpler convolutional pipelines.

Compared to prior work, Wang et al. [20] employed transfer learning with AlexNet and VGG-16, reporting moderate accuracy levels but facing limitations due to the relatively shallow and less efficient architectures. Deepa et al. [23] explored ensemble methods combining InceptionV3 and Xception, achieving improved results at the cost of significantly higher computational overhead. In contrast, our use of DenseNet-121 provides a better compromise between accuracy, efficiency, and generalizability, particularly when evaluated across heterogeneous datasets. Fig. 7 illustrates the training and validation accuracy and loss across the Private, APTOS, and EyePACS datasets, where consistently high accuracy and steadily decreasing loss confirm the robustness and transferability of DenseNet-121 across different data sources.

The confusion matrix is a valuable tool that provides crucial insights into the performance of a model, enabling the evaluation of its accuracy in classifying instances of DR and No DR. The assessment of classification results for the DR and No DR categories was conducted across three distinct datasets using the confusion matrix. However, Table 6 illustrates the confusion matrix for each dataset, highlighting the consistently superior performance of the DenseNet-121 architecture. Considering these findings, DenseNet-121 has been selected as the optimal architecture due to its exceptional predictive capabilities.

Other performance indices for the transfer learning models are shown in Table 7, which offers a comprehensive overview of the performances of DenseNet-121, MobileNet-v2, and Inception-v3. Across the three evaluated datasets (Private, APTOS, and EyePACS). DenseNet-121 consistently demonstrates remarkable performance with high AUROC values for each dataset. For example, on the Private dataset, DenseNet-121 achieves an impressive AUROC of 0.9991, whereas MobileNet-v2 and Inception-v3 exhibit competitive but slower AUROC values of 0.9974 and 0.9945, respectively. These values highlight the efficacy of the models in accurately classifying the datasets. Furthermore, the reported inference times, such as 161.62 ms for DenseNet-121 on the Private dataset, provide insights into the computational efficiency of the models.

Fig. 8 visualizes the AUROC values obtained by DenseNet-121, MobileNet-v2, and Inception-v3 across different datasets. DenseNet-121 consistently achieves the highest AUROC values, demonstrating its superior performance in classification tasks compared to the other models. Although MobileNet-v2 and Inception-v3 exhibit slightly lower AUROC values, they still showcase competitive performance. These findings provide a comprehensive understanding of the performance and efficiency trade-offs among the evaluated models. DenseNet-121 stands out as the top performer in terms of AUROC values, indicating its effectiveness in classification tasks. On the other hand, MobileNet-v2 and Inception-v3 offer computational advantages in terms of faster inference times due to their lighter and less complex architectures. These insights are crucial for selecting the most appropriate transfer

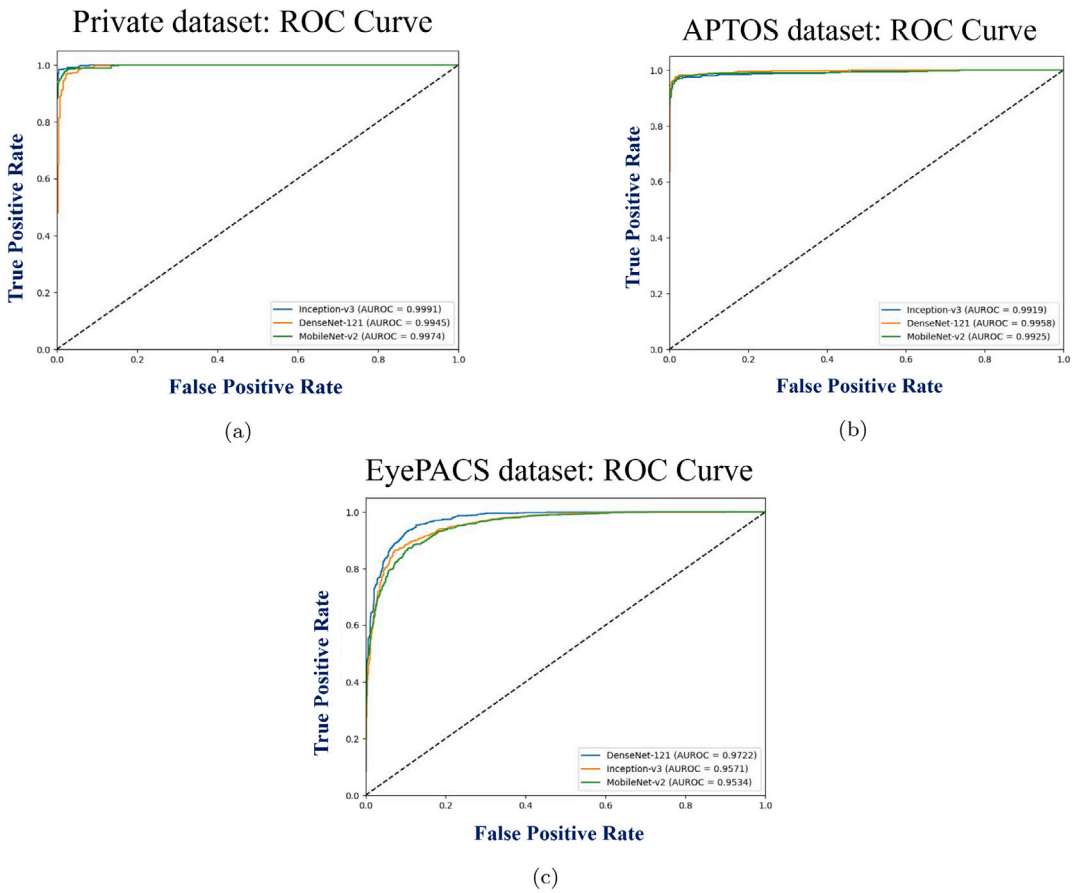


Fig. 8. ROC curves obtained from models fitted to the Private (a), APTOS (b), and EyePACS (c) datasets.

Table 7
Summary of performance indices for benchmark analysis: Evaluating model complexity and inference time.

Dataset	Model	Model depth	Model complexity	Memory size (MB)	AUROC	Inference time (ms)
Private	DenseNet-121	431	**32.72**	**223.99**	**0.9991**	161.62
	MobileNet-v2	**158**	**34.37**	254.09	0.9974	**118.65**
	Inception-v3	315	48.01	284.02	0.9945	133.50
APTOS	DenseNet-121	431	**32.72**	**223.99**	**0.9958**	210.03
	MobileNet-v2	**158**	**34.37**	254.09	0.9925	**157.93**
	Inception-v3	315	48.01	284.02	0.9919	170.84
EyePACS	DenseNet-121	431	**32.72**	**223.99**	**0.9722**	137.52
	MobileNet-v2	**158**	**34.37**	254.09	0.9534	**104.72**
	Inception-v3	315	48.01	284.02	0.9571	105.54

learning model based on the specific requirements of the application at hand.

The researchers explored different transfer learning models in the field of DR detection [18–24]. In the end, Table 8 presents a detailed comparison of our approach with the state of the art (SOTA), highlighting our ability to improve the quality of retinal images captured by mobile devices or fundus retinography. This improvement in image quality is essential for accurate detection of DR, offering the potential to prevent blindness, particularly in older patients.

4.6. Real-time mobile app-based tele ophthalmology

The development and deployment of a real-time diagnostic system for DR include several stages. Firstly, retinal images are captured in real-time using an oDocs nun IR handheld fundus camera. Next, the retinal images are enhanced using the specified algorithms. The optimized deep learning models, specifically the DenseNet-121 model, were optimized and converted into TensorFlow Lite format for deployment on mobile devices. The model was then deployed on smartphones, with the classification process, including preprocessing and prediction, executing with an average inference time of 1.2 s per image. This performance is suitable for real-time analysis in mobile screening environments where immediate feedback is required during patient examinations. The timing was measured using Python's time module across 100 test images. This demonstrates that the model can perform real-time inference on smartphones, meeting the practical requirements of mobile DR screening systems [29]. Then, a DR prediction DenseNet-121 model is adapted and fine-tuned to accurately identify the presence or absence of DR pathology. The DenseNet-121 model was adapted and fine-tuned, selected from several alternative models. Then, it is deployed on mobile smartphone applications. This accessibility enables specialist ophthalmologists to use the mobile App effectively to classify and diagnose DR. Diagnostic results are transmitted in real time to a pilot site affiliated with the Faculty of Medicine at the HASSAN II Hospital in Fez, enabling centralized monitoring and regular assessment. This integrated system provides crucial clinical support and

Table 8
A comparison with state-of-the-art methods.

Works	Technique	Dataset	Performance metrics
Gulshan et al. [18]	CNNs	EyePACS	Sen = 90.3% , Spe = 98.1%
Wang et al. [20]	TL AlexNet, VGG-16 and InceptionNetV3	EyePACS	Acc = 37.43%, 50.03%, 63.23%
Pamadi et al. [21]	TL MobileNetV2	APTOS	Acc = 78%
Zhou et al. [22]	Multi-Task learning	EyePACS	Acc = 63.22%
Deepa et al. [23]	Ensemble method (InceptionV3, Xception)	EyePACS	Acc = 96.2% for Ensemble method, Acc = 85.4% for InceptionV3
Qureshi et al. [24]	Active Deep Learning (ADL)	EyePACS	Acc = 95.10%, Sen = 92.20%, Sep = 95.10%
Our approach	**Fine-tunned DenseNet-121**	**Private, EyePACS and APTOS**	**Acc = 98.61%, 98.03%, 90.90%; Sen = 98.33%, 97.76%, 93.90%; Sep = 100.00%, 99.78%, 99.40%**

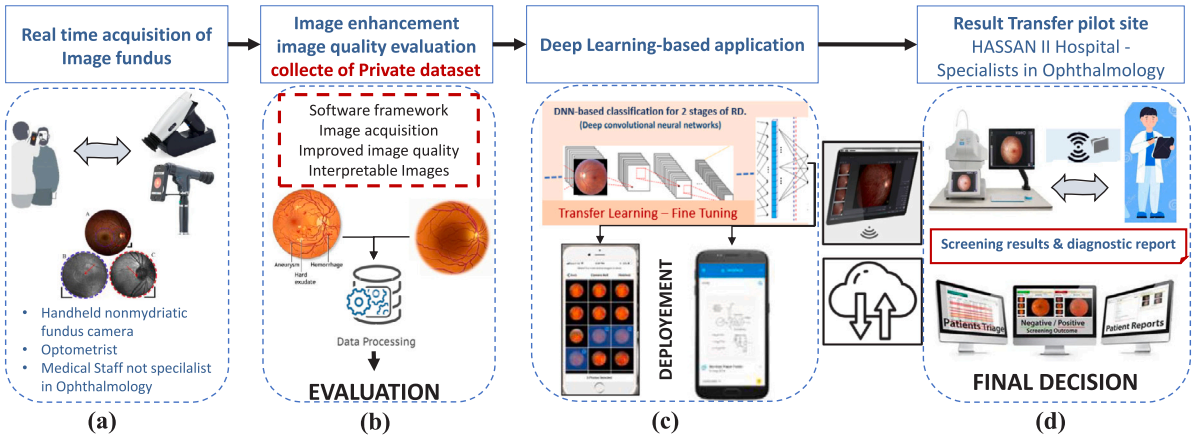


Fig. 9. The early DR screening in remote medical centers or resource-limited areas: A comprehensive workflow from real-time acquisition (a), Image enhancement (b), Deep learning Model application (c), to Screening & transfer result (d).

improves medical decision-making, enabling early detection and more effective management of DR. Fig. 9 presents the protocol of earlier screening of DR in remote medical centers or areas with limited access to ophthalmologists.

The mobile system interfaces with digital tablets and desktop applications through a cloud-based server. This server allows for real-time data synchronization across platforms, ensuring that diagnostic results from the mobile application are instantly available on both tablet and desktop interfaces. By implementing this cloud-based communication, we ensure that the system can be effectively used in various healthcare settings, from remote areas with limited infrastructure to more equipped hospital environments. Furthermore, the tablet and desktop applications will offer enhanced features for medical professionals, such as data visualization and patient management system integration, making the system versatile for both mobile and stationary use.

The portability of the proposed system significantly enhances its applicability in resource-limited settings, where access to conventional fundus imaging infrastructure is restricted. By relying on a compact handheld retinal camera and a lightweight mobile application, the system can be deployed in rural or semi-urban areas where patients may not have access to regular ophthalmological consultations. Despite challenges such as unstable lighting and patient movement, the system maintains robust diagnostic performance due to its adaptive image enhancement pipeline. Furthermore, real-time processing on smartphones ensures immediate feedback, supporting faster decision-making and timely referral. This portability bridges the gap between underserved communities and early diabetic retinopathy detection, contributing to more equitable healthcare delivery.

The DenseNet-121 model, recognized for its strong performance, is integrated into a mobile application offering a practical and portable solution. A comprehensive planning process was employed to delineate the application's objectives, analyze the distinct requirements of end-users, particularly healthcare practitioners operating in clinical settings such as medical caravans, and establish the core functional specifications. The technical implementation leverages modern technologies, including JavaScript as the programming language, to ensure smooth and efficient integration of the DenseNet-121 model. The application facilitates optimized image processing and accurate automated diagnosis. This interface minimizes user effort, making interactions more efficient in time-sensitive, resource-constrained settings. Key features like real-time image acquisition, enhanced image quality using advanced techniques, and user-friendly result visualization have been integrated. Additionally, note that, for the time being, the system has only been tested by us and a few members of the medical staff. In the next stages, once it has been validated by the authorities, it will be made available to other users, who are not necessarily familiar with it. At that point, we will gather their feedback and further evaluate their performance. This study will soon be initiated by our team, along with other users, and only then will we be able to truly assess the user experience. This will be the subject of future studies.

Fig. 10 provides a detailed description of the deployment steps for the DenseNet-121 optimized and fine-tuned model on a mobile app as

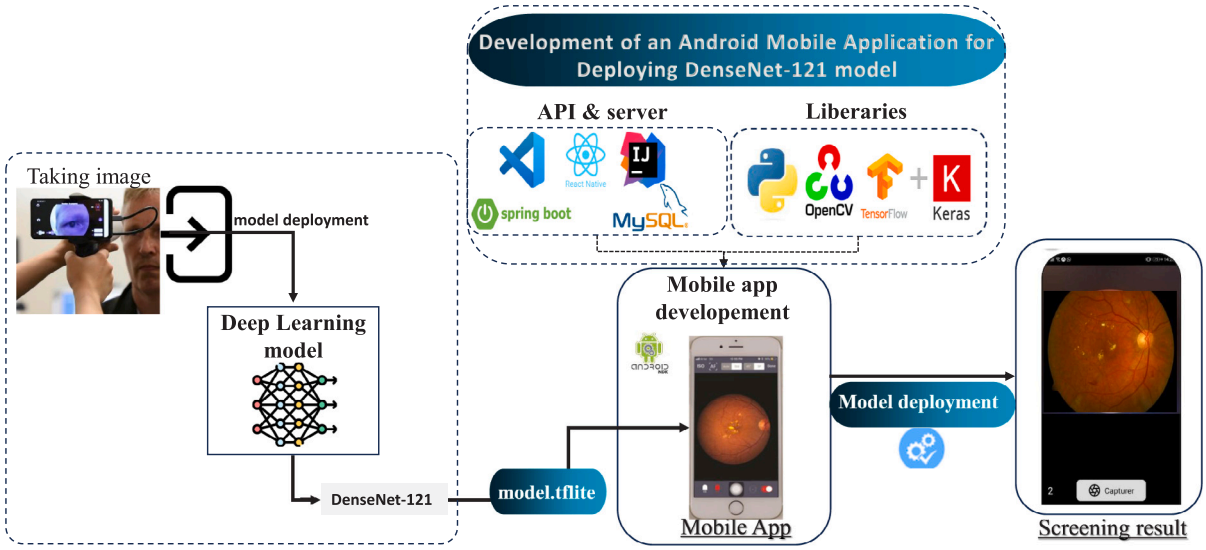


Fig. 10. The workflow for implementation on smartphone, desktop, and tablet platforms.

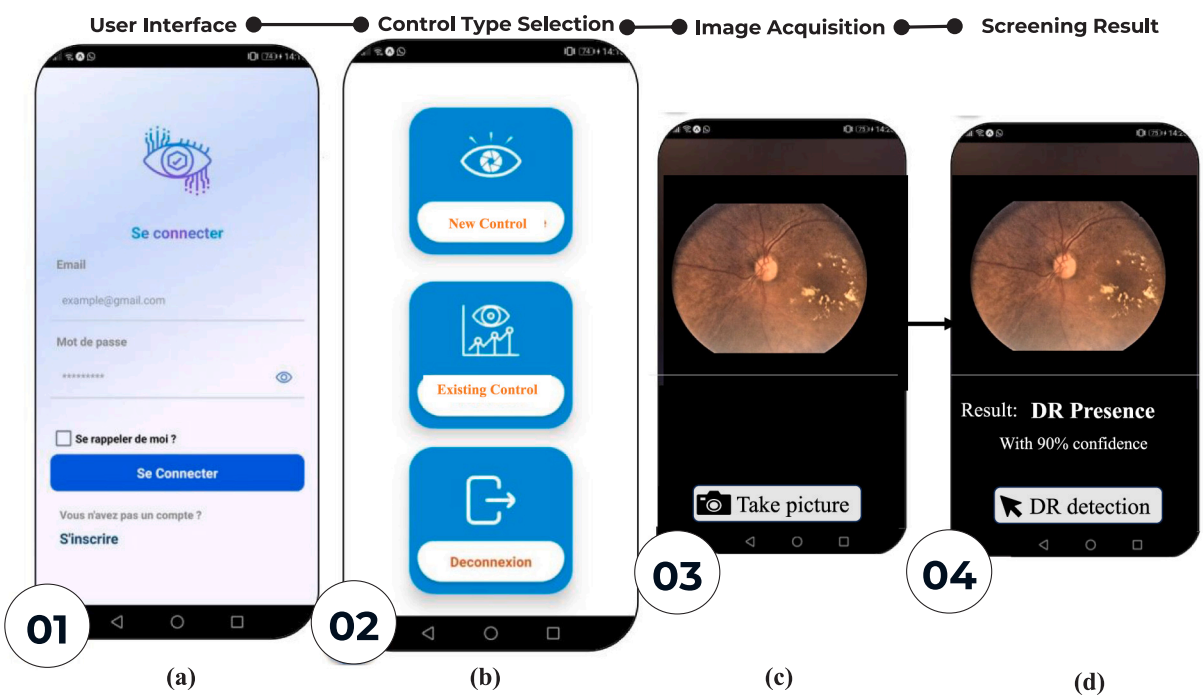


Fig. 11. Demonstration of smartphone graphical interface for DR detection: User interface for connection (a), Control type selection (b), Image acquisition (c), and Screening result display.

an initial prototype. The conversion of the model to a *.tflite* format is facilitated using the TensorFlow Lite interpreter. Subsequently, the *.tflite* model is deployed on smartphones using the Java programming language within the Android Studio Integrated Development Environment (IDE). Several libraries, including TensorFlow, Keras, and OpenCV, are utilized during the implementation to ensure efficient image processing and smooth execution of the model. The successful deployment of the model on smartphones offers a practical and effective solution for the early detection of DR.

The implementation on smartphones provides several advantages, notably the high portability and accessibility of screening, which eliminates the need for specialized equipment. These features are particularly beneficial in settings where medical resources are scarce, such as rural areas or during medical caravans. Additionally, extending the solution to other platforms, such as desktops and tablets, further enhances its versatility and overall impact. Fig. 11 demonstrates the successful deployment of the model on an Android smartphone, showcasing the system's ability to deliver real-time, accurate results. Moreover, our vision extends beyond DR detection. Once the system is established and validated for DR diagnosis, the goal is to generalize the model and broaden its scope to detect other ocular pathologies, such as glaucoma, age-related macular degeneration (AMD), and more. This expansion will include training the model on diverse datasets specific to each pathology, ultimately empowering clinicians to use the application as a versatile tool for diagnosing a wide range of ocular conditions.

In addition, the combination of two publicly available datasets, APTOS and EyePACS, with our private dataset has validated the results obtained. The integration of multiple diverse datasets reduces the risk of bias that may result from dependence on a single data source. The

APTOS and EyePACS datasets contribute valuable variability, encompassing a broad spectrum of patient demographics, imaging conditions, and resolution qualities, enhancing the model's generalizability. Moreover, by incorporating our private dataset, we introduce real-world data specific to the population at CHU Hassan II, ensuring that the model is better suited for deployment in clinical settings with similar demographic characteristics. This diversified approach ultimately strengthens the robustness and reliability of the model, enabling it to perform effectively across various scenarios and environments and supporting its use for real-time, early screening in mobile applications.

Ultimately, to make our approach more generalized, some further work is still needed, in particular, we need to integrate more images into our database, even captured by multiple cameras, and incorporate more diversity in terms of age, gender, and regions. In addition, user feedback needs to be analyzed to make the system ergonomic, easy to use, and more interactive. Future work will also focus on integrating uncertainty estimation and extending the system's validation to different fundus cameras, populations, and clinical conditions to improve robustness and practical applicability.

5. Conclusions and future works

In this paper, we proposed a Deep Convolutional Neural Network (DCNN) model based on a transfer learning approach and fine-tuning technique for the early detection of diabetic retinopathy (DR). Our approach utilizes an optimized and fine-tuned DenseNet-121 architecture and demonstrates high accuracy on both public (APTOS and EyePACS) and private datasets. The model achieved classification accuracies of 98.61% on the private dataset, 98.03% on APTOS, and 90.90% on EyePACS. These results confirm the robustness and generalization capability of the proposed method. Our main contributions include: (i) the development and optimization of a lightweight and high-performing DCNN model for DR classification; (ii) the integration of the model into a mobile application for real-time inference using a handheld fundus camera; and (iii) an extensive evaluation on multiple datasets to ensure adaptability and consistency across different image sources.

Our work contributes to the field of medical image analysis by demonstrating the effectiveness of transfer learning across multiple datasets, while also showing that deep learning models can be efficiently deployed in resource-constrained environments. Practically, the proposed mobile screening solution is portable, cost-effective, and capable of real-time inference (161.62 ms per image), making it well-suited for use in underserved areas. This enables early DR detection in regions with limited access to specialists, potentially improving clinical outcomes and reducing preventable vision loss. As future work, we plan to improve the efficiency of the model by reducing inference time further and extending the deployment to tablets and desktop systems. We also aim to conduct a field validation campaign in the Fez region in Morocco to evaluate the system in real-world clinical settings. At the end, we also look forward to expanding our dataset and analyzing the users' experience to make our system more intuitive, ergonomic, and easier to use.

CRediT authorship contribution statement

El-Mehdi Chakour: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Zineb Sadok:** Writing – original draft, Methodology, Investigation, Conceptualization. **Rostom Kachouri:** Validation, Supervision, Methodology, Investigation, Formal analysis. **Anass Mansouri:** Visualization, Validation, Supervision, Software, Resources, Methodology, Formal analysis, Conceptualization. **Idriss Benatiya Andaloussi:** Visualization, Validation, Resources, Formal analysis, Data curation. **Mohamed Akil:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation. **Ali Ahaitouf:** Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

APTOS dataset is available at www.kaggle.com/competitions/aptos2019-blindness-detection. EyePACS dataset is available at www.kaggle.com/competitions/diabetic-retinopathy-detection/data. The private dataset is not available for public access.

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